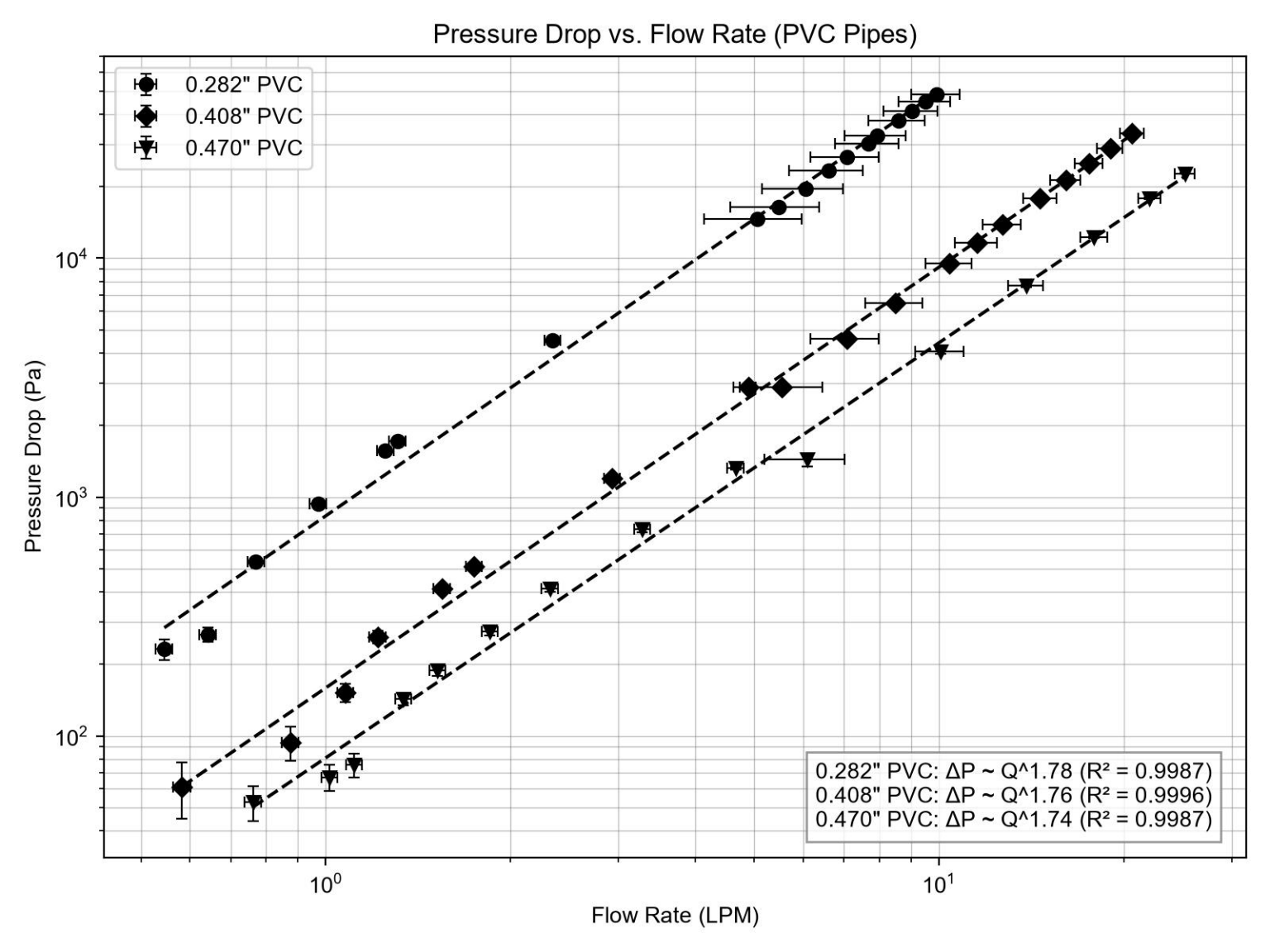
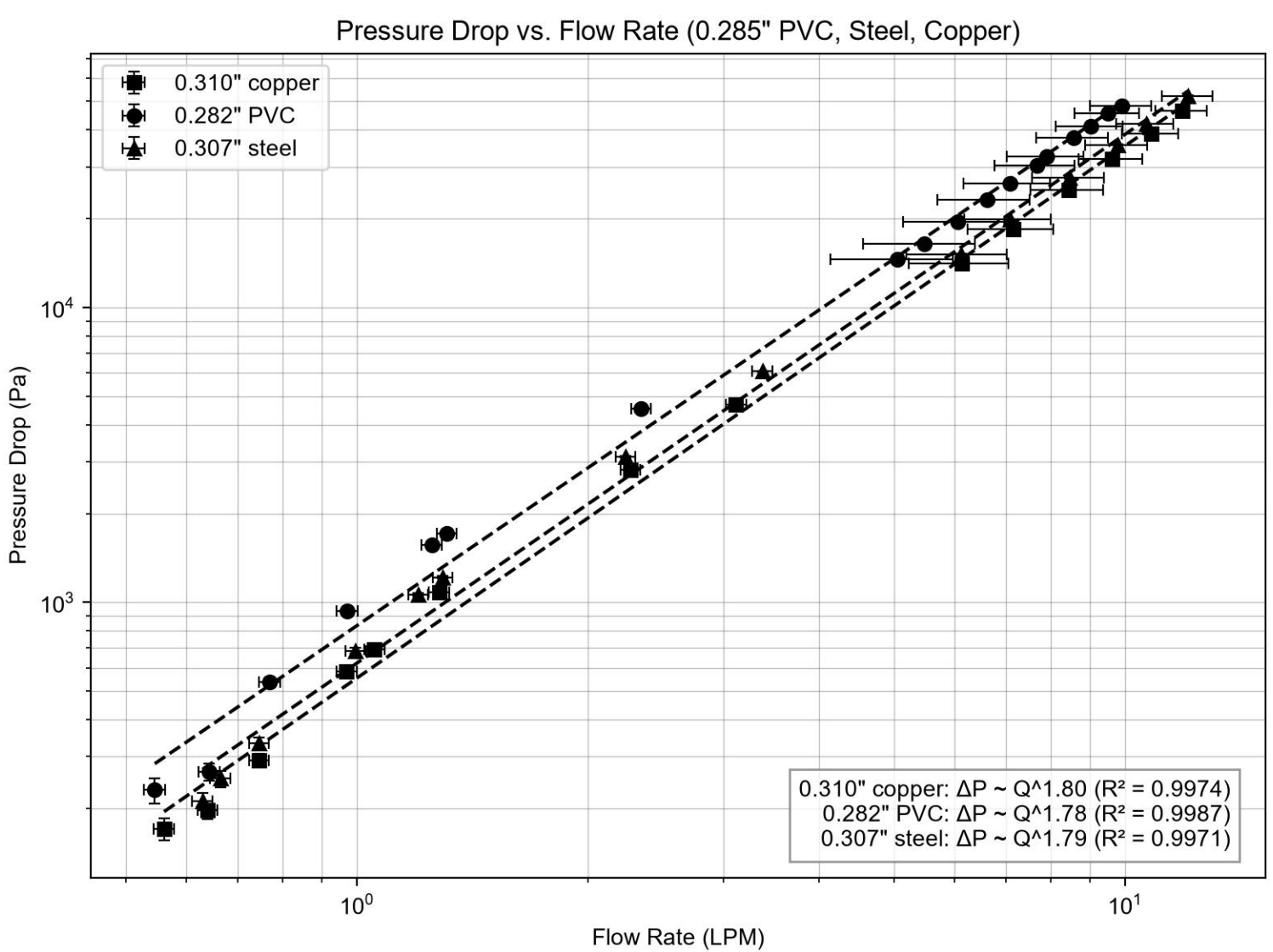
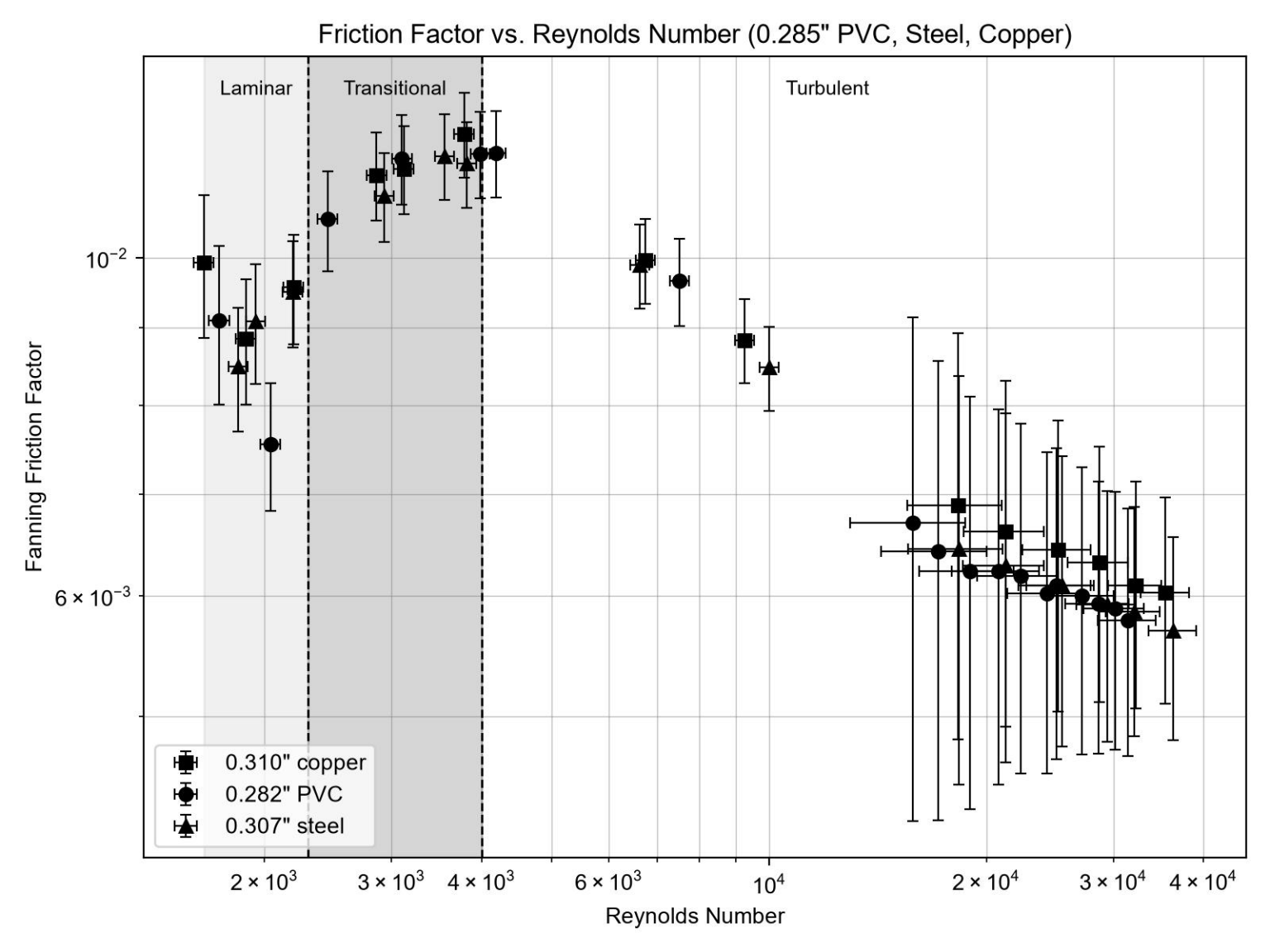
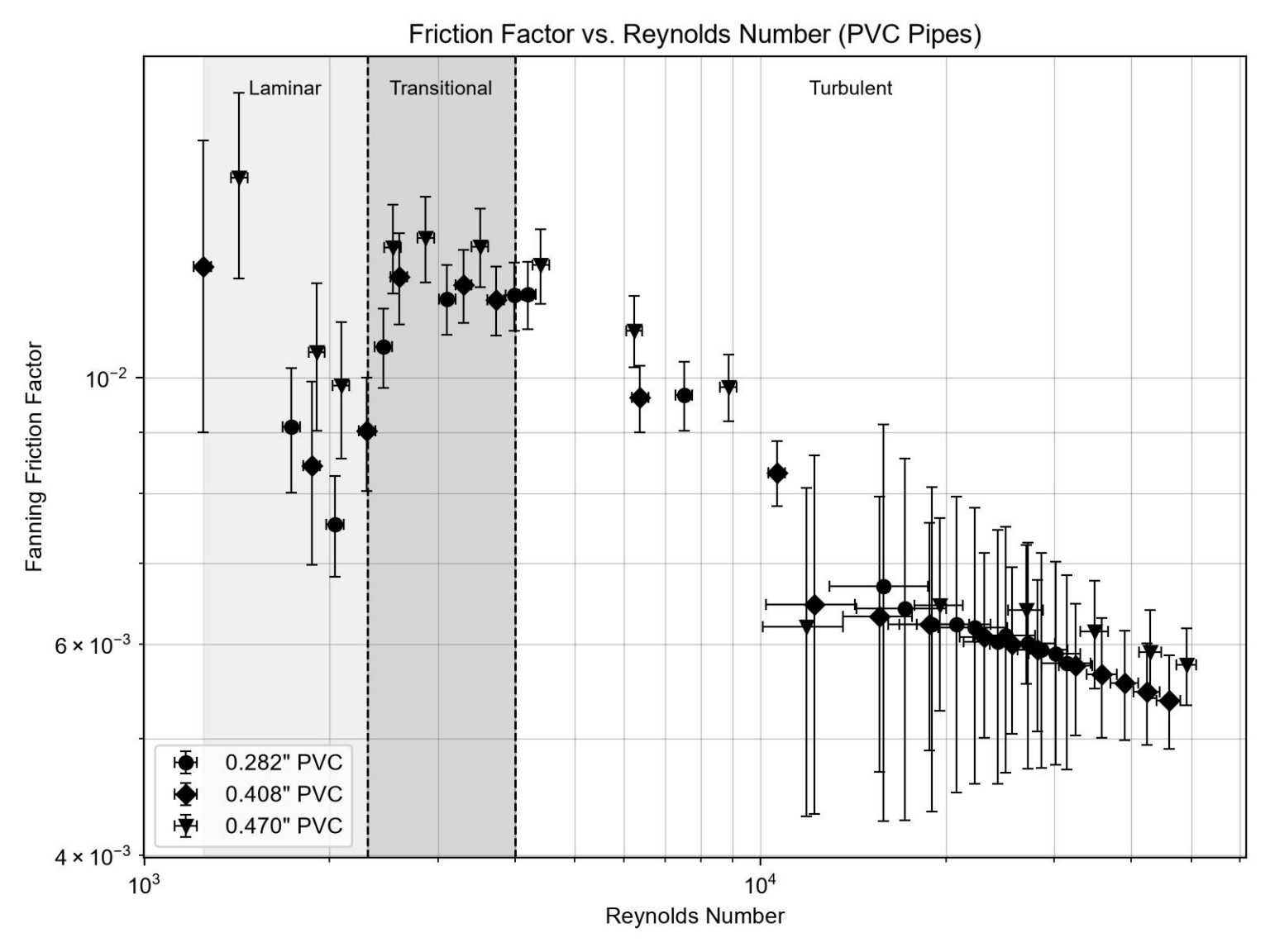
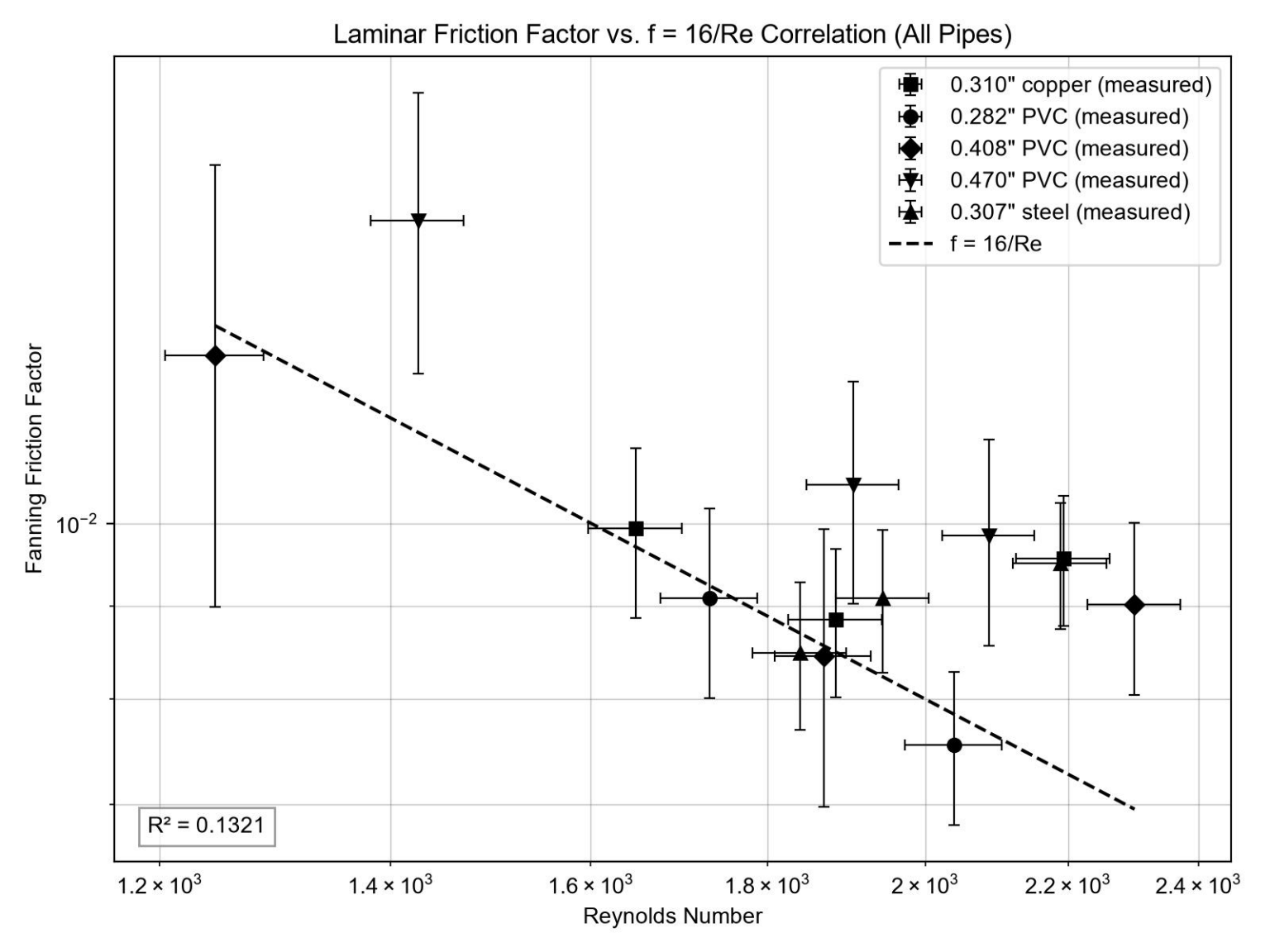
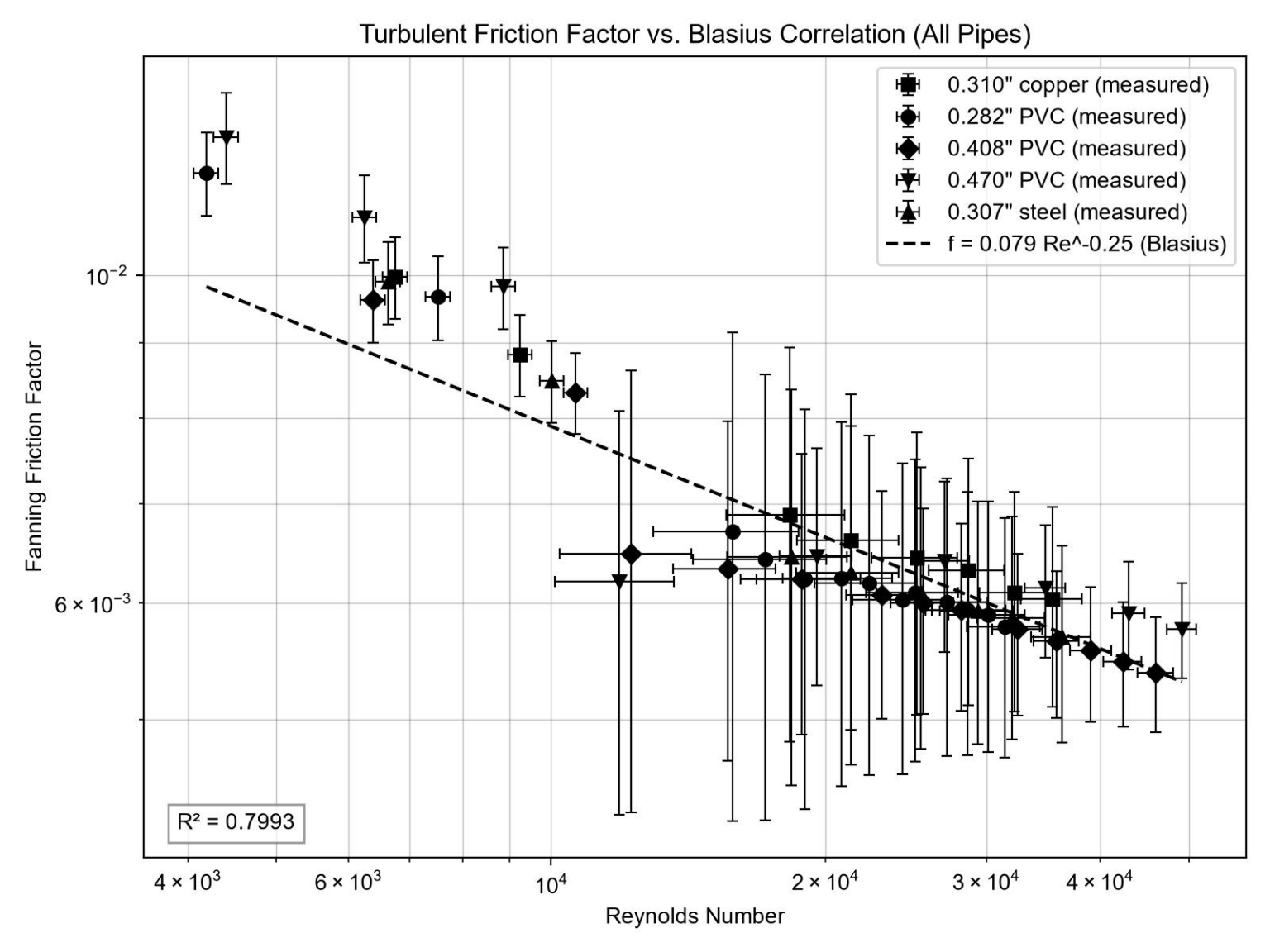

Pressure Drop vs. Flow Rate (PVC Pipes)
0.282" PVC
0.408" PVC
0.470" PVC
Pressure Drop (Pa)
Flow Rate (LPM)
10^2
10^3
10^4
10^0
10^1
0.282" PVC: ΔP ~ Q^1.78 (R² = 0.9987)
0.408" PVC: ΔP ~ Q^1.76 (R² = 0.9996)
0.470" PVC: ΔP ~ Q^1.74 (R² = 0.9987)

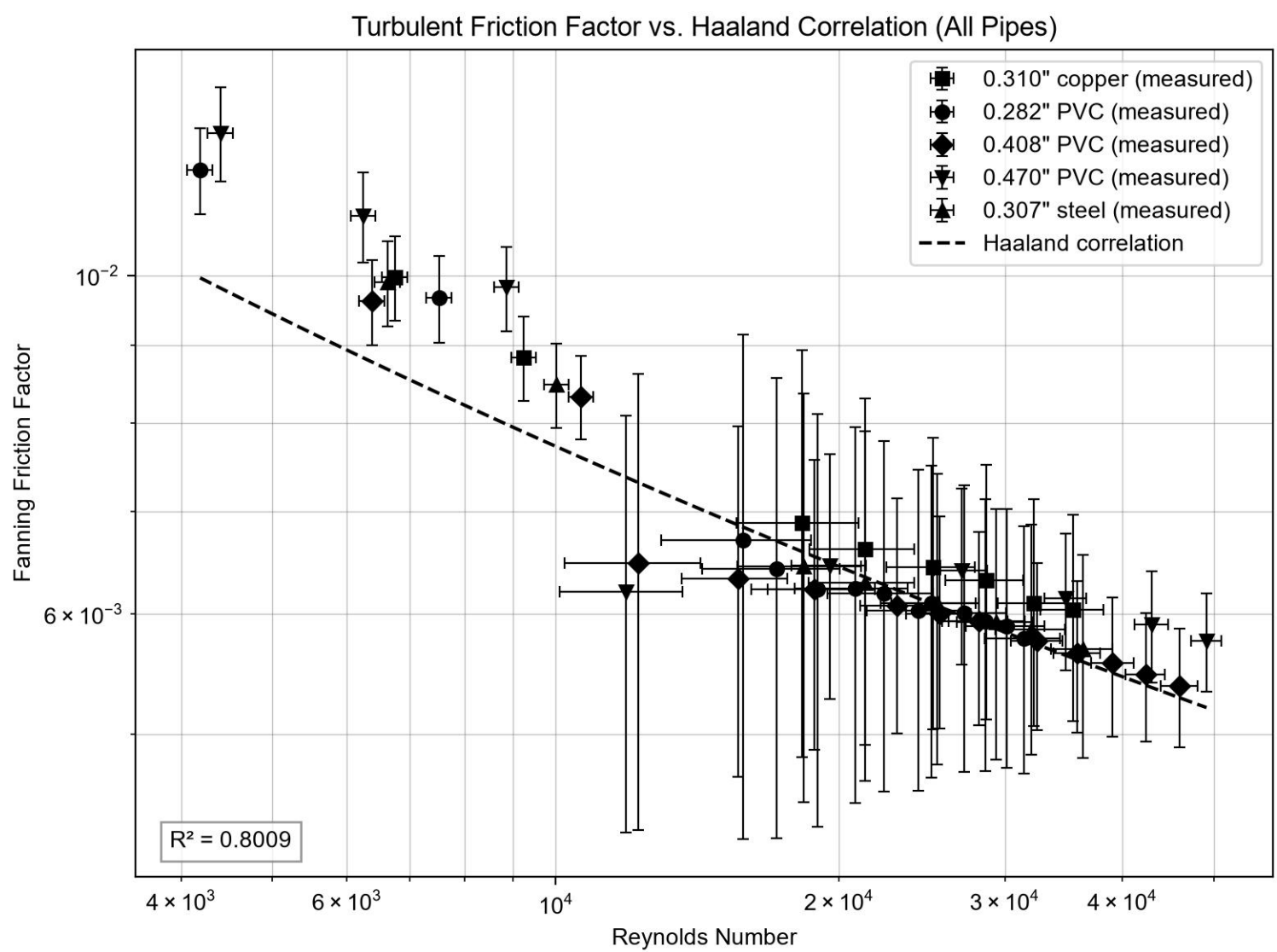

Pressure Drop vs. Flow Rate (0.285" PVC, Steel, Copper)
0.310" copper
0.282" PVC
0.307" steel
Pressure Drop (Pa)
Flow Rate (LPM)
10^4
10^3
10^0
10^1
0.310" copper: ΔP ~ Q^1.80 (R² = 0.9974)
0.282" PVC: ΔP ~ Q^1.78 (R² = 0.9987)
0.307" steel: ΔP ~ Q^1.79 (R² = 0.9971)

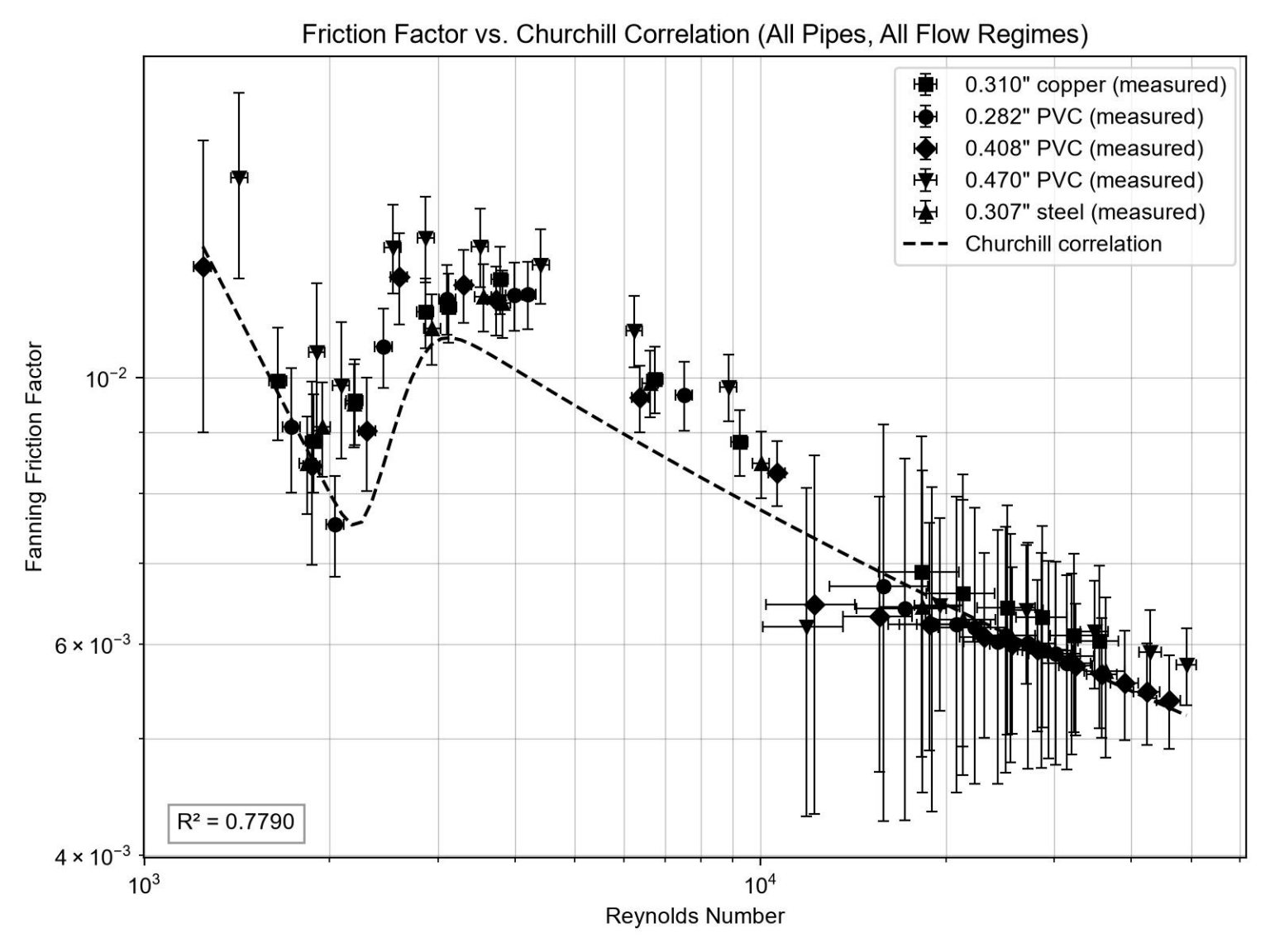

Friction Factor vs. Reynolds Number (0.285" PVC, Steel, Copper)
Laminar
Transitional
Turbulent
Fanning Friction Factor
10^{-2}
6×10^{-3}
0.310" copper
0.282" PVC
0.307" steel
2×10^3
3×10^3
4×10^3
6×10^3
10^4
2×10^4
3×10^4
4×10^4
Reynolds Number

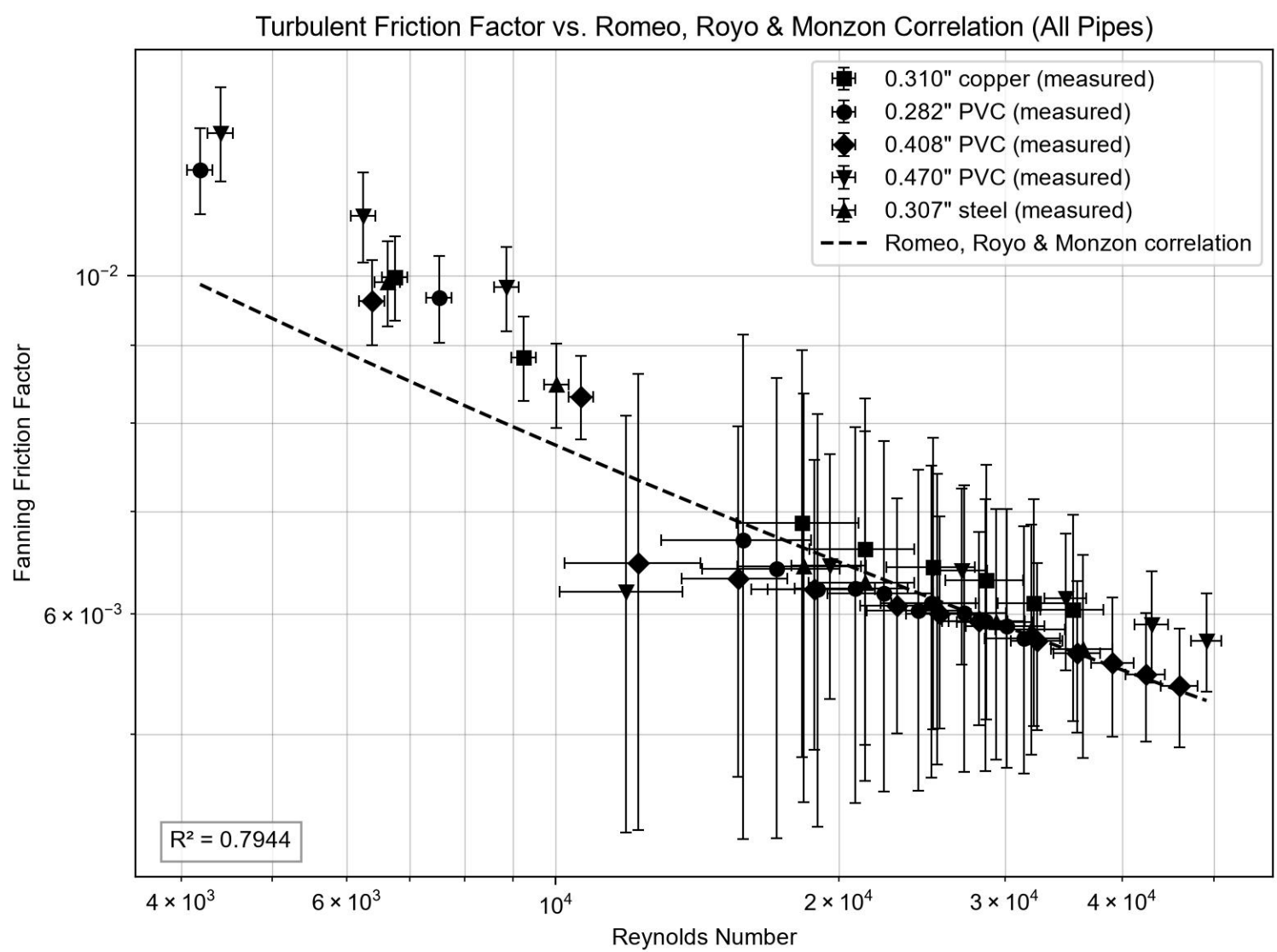

Friction Factor vs. Reynolds Number (PVC Pipes)
Laminar
Transitional
Turbulent
Fanning Friction Factor
10^{-2}
6×10^{-3}
4×10^{-3}
10^{3}
10^{4}
Reynolds Number
0.282" PVC
0.408" PVC
0.470" PVC

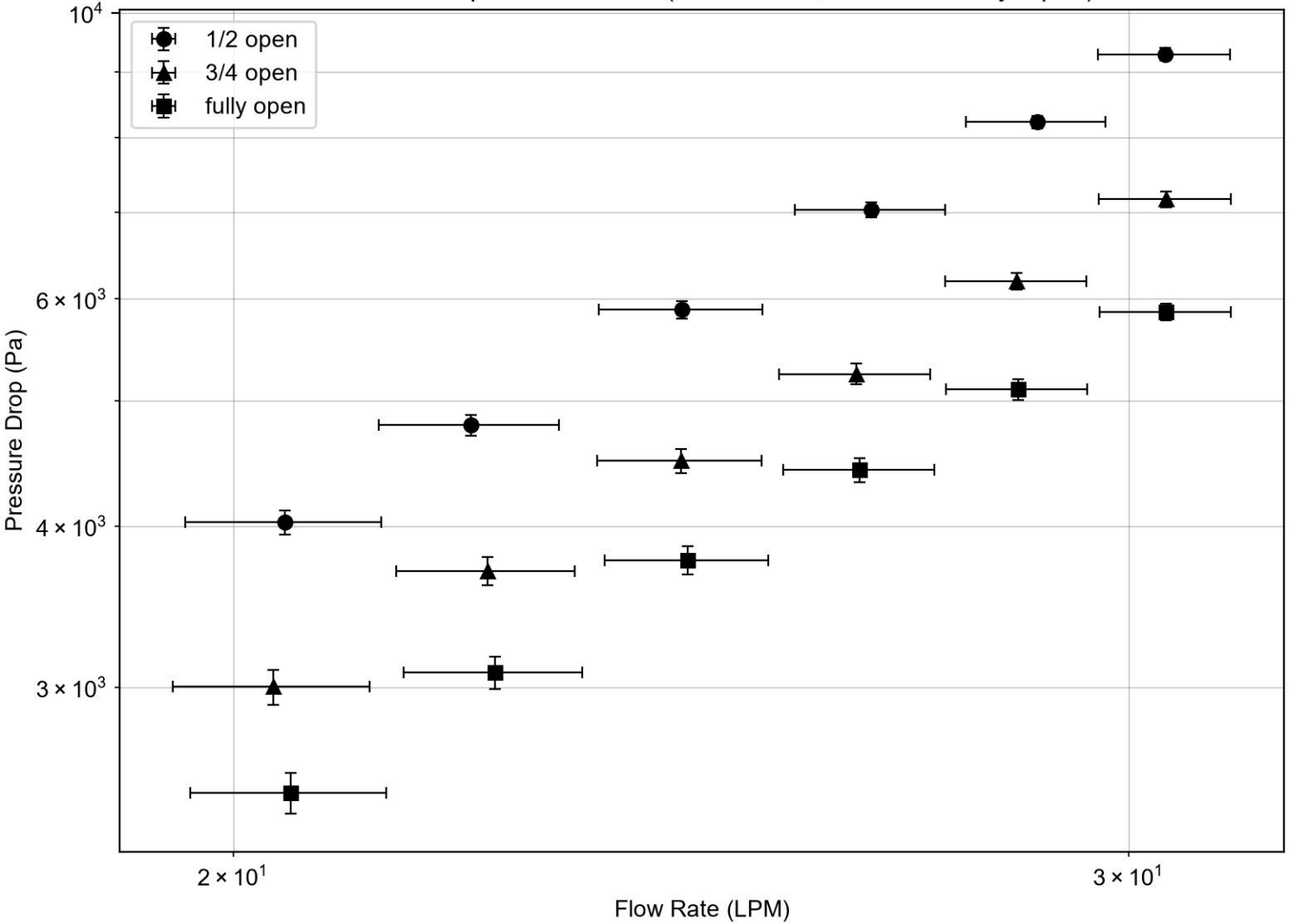

Laminar Friction Factor vs. f = 16/Re Correlation (All Pipes)
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
f = 16/Re
R² = 0.1321
Fanning Friction Factor
10^{-2}
1.2×10^3
1.4×10^3
1.6×10^3
1.8×10^3
2×10^3
2.2×10^3
2.4×10^3
Reynolds Number

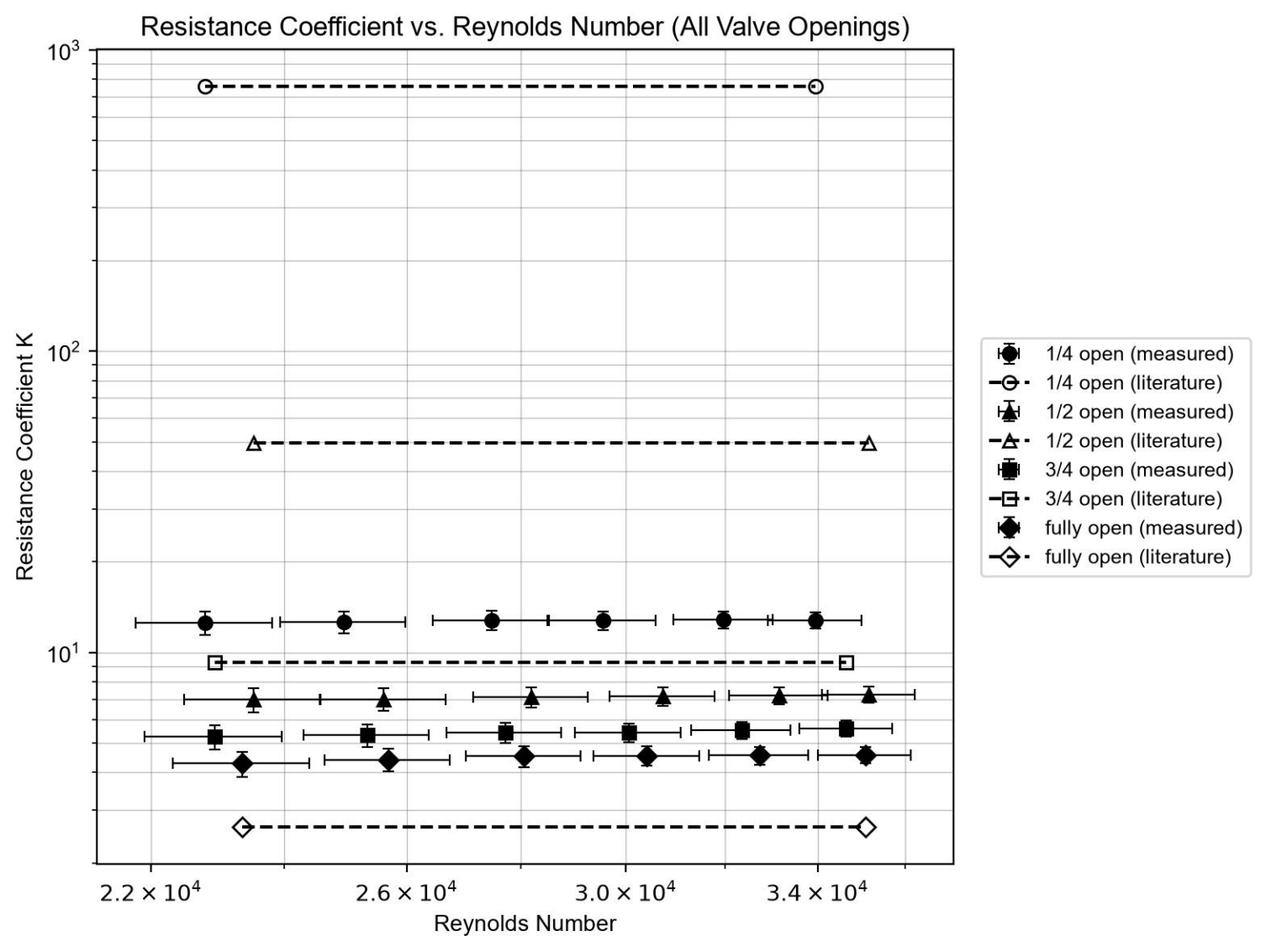

Turbulent Friction Factor vs. Blasius Correlation (All Pipes)
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
f = 0.079 Re^-0.25 (Blasius)
R² = 0.7993
Fanning Friction Factor
Reynolds Number
10⁻²
6 × 10⁻³
4 × 10³
6 × 10³
10⁴
2 × 10⁴
3 × 10⁴
4 × 10⁴

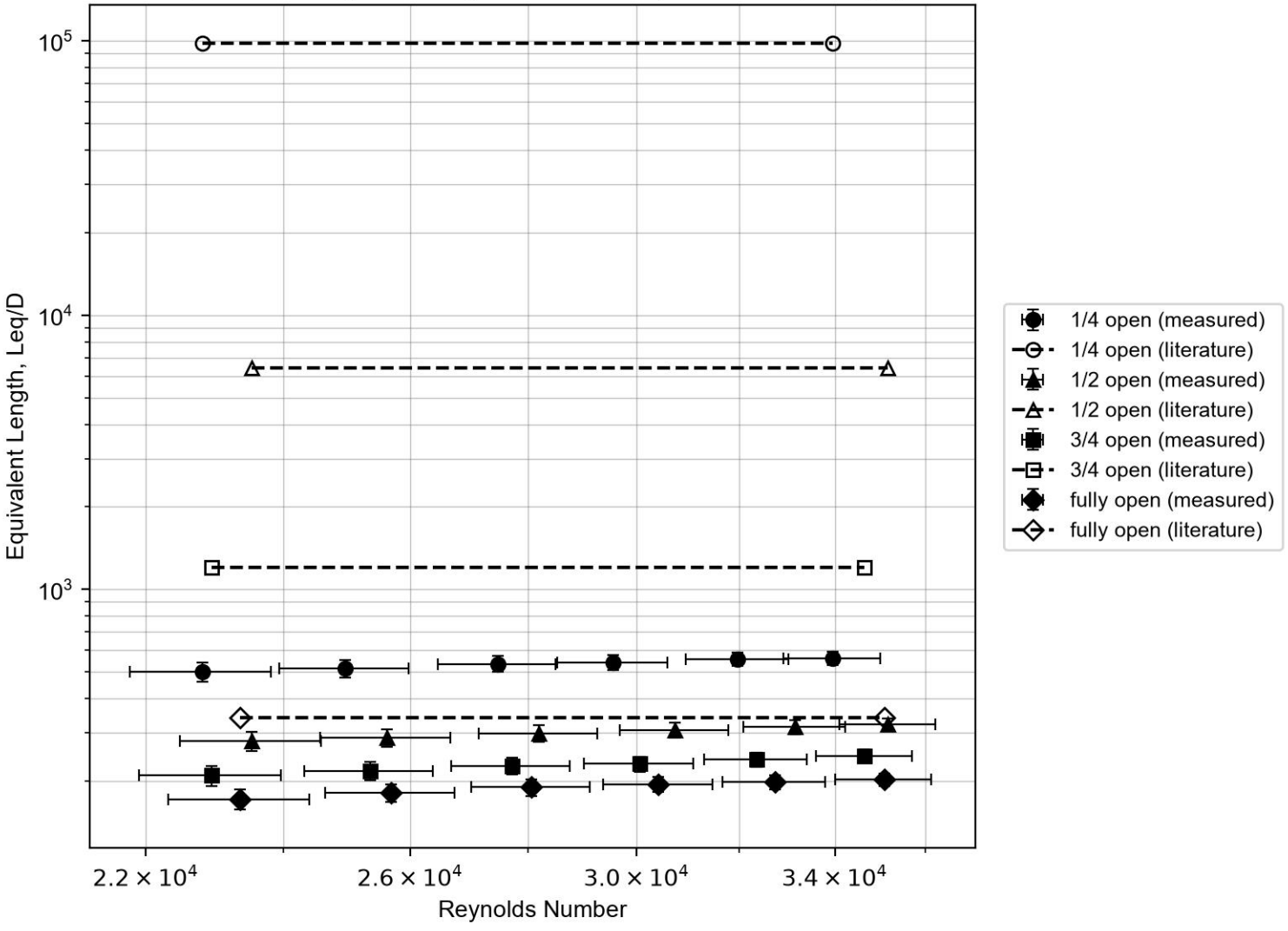

Turbulent Friction Factor vs. Haaland Correlation (All Pipes)
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
Haaland correlation
Fanning Friction Factor
10^{-2}
6×10^{-3}
$R^2 = 0.8009$
4×10^3
6×10^3
10^4
2×10^4
3×10^4
4×10^4
Reynolds Number


Friction Factor vs. Churchill Correlation (All Pipes, All Flow Regimes)
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
Churchill correlation
Fanning Friction Factor
10^{-2}
6×10^{-3}
4×10^{-3}
10^{3}
10^{4}
Reynolds Number
R² = 0.7790


Turbulent Friction Factor vs. Romeo, Royo & Monzon Correlation (All Pipes)
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
Romeo, Royo & Monzon correlation
Fanning Friction Factor
10^{-2}
6×10^{-3}
$R^2 = 0.7944$
4×10^3
6×10^3
10^4
2×10^4
3×10^4
4×10^4
Reynolds Number


Pressure Drop vs. Flow Rate (Globe Valve, 1/2 / 3/4 / Fully Open)
1/2 open
3/4 open
fully open
10^4
6×10^3
4×10^3
3×10^3
Pressure Drop (Pa)
2×10^1
3×10^1
Flow Rate (LPM)


Resistance Coefficient vs. Reynolds Number (All Valve Openings)
Resistance Coefficient K
10^3
10^2
10^1
2.2×10^4
2.6×10^4
3.0×10^4
3.4×10^4
Reynolds Number
1/4 open (measured)
1/4 open (literature)
1/2 open (measured)
1/2 open (literature)
3/4 open (measured)
3/4 open (literature)
fully open (measured)
fully open (literature)


Equivalent Length vs. Reynolds Number (All Valve Openings)
Equivalent Length, Leq/D
10^5
10^4
10^3
2.2×10^4
2.6×10^4
3.0×10^4
3.4×10^4
Reynolds Number
1/4 open (measured)
1/4 open (literature)
1/2 open (measured)
1/2 open (literature)
3/4 open (measured)
3/4 open (literature)
fully open (measured)
fully open (literature)